

SIH 2026 – Problem Statement 26005

MASTER PROJECT KNOWLEDGE DOCUMENT

Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in the North Eastern Region (NER)

Stages 1–10 • Research • Engineering Decisions • Architecture • Electronics • Testing Plan

Document item	Value
Problem Statement	SIH 2026 PS 26005
Category	Hardware / Agriculture / FoodTech / Rural Development
Target context	Decentralized short-term cold storage at village collection centres / FPOs / cooperatives / farm-gate points in NER
Current design capacity	≈ 200 kg produce
Primary prototype crop	Tomato; architecture intended to support compatible moderate-temperature vegetables
Current target chamber temperature	≈ 10°C nominal; adjustable rather than permanently fixed
Current thermal-backup concept	≈ 25 kg PCM, recorded as IP08/ZP08 in the handwritten notes; exact commercial material/specification requires final verification
Document status	Engineering concept / pre-prototype master record — physical validation still required
Primary source supplied by team	SIH 26005 STAGE 1-10 20260918.pdf — 40 handwritten pages

This document is a structured digital consolidation of the team's handwritten 40-page record and the project decisions developed during the SIH work. It intentionally distinguishes evidence, calculations, engineering assumptions and prototype targets. Where the handwritten record contains conflicting or ambiguous values, the conflict is preserved and flagged instead of being silently corrected.

1. How This Document Should Be Used

This is the team's single working context for SIH 26005. It is intended to be shared with every team member and can also be uploaded to an AI assistant before asking for project-specific help.

- Do not restart the project from zero unless a new requirement or new evidence requires it.
- Treat the decisions in this document as the current design baseline, not as experimentally proven final specifications.
- Every important number should be tagged as Evidence-Based, Calculated, Engineering Assumption, or Prototype Target.
- If a new proposal changes a major subsystem, compare it against the current architecture and explain what existing decision it replaces.
- Do not convert a target into a verified performance claim without a test.
- For refrigeration work involving R290, charging and sealed-system work must be performed by a qualified refrigeration technician; the team should not improvise refrigerant charging.
- Stage 9 contains validation tests that are planned, not completed, unless a later test record is added.

2. Evidence / Decision Classification

Label	Meaning	Example in this project
Evidence-Based	Supported by external research, datasheets, standards, or documented observations.	Vegetable storage conditions, manufacturer PCM data, equipment datasheet values.
Calculated	Derived mathematically from stated assumptions or measurements.	$200\text{ kg} \times \text{specific heat} \times \text{temperature difference}$.
Engineering Assumption	Chosen because a design needs a working value before physical testing.	Incoming produce temperature $\approx 30^\circ\text{C}$; 100 mm PUF insulation in the initial design.
Prototype Target	A value the prototype is intended to achieve and must later be tested.	$\approx 10^\circ\text{C}$ chamber target, 6–12 h thermal-backup target, temperature uniformity.

3. Project Concept in One Page

The project evolved from a simple short-duration Peltier/battery cooling idea into a solar-powered controlled cold-storage system using vapour-compression refrigeration plus PCM thermal storage. The refrigeration system creates the cold; PCM stores thermal capacity so the chamber can remain more stable during night-time operation or refrigeration interruption; insulation reduces heat ingress; the battery stores electrical energy; solar supplies energy; and the ESP32 coordinates sensing, control, alerts and monitoring.

Function	Subsystem	Role
Supply energy	Solar PV	Generate electrical energy during daylight.
Store electrical energy	48 V, 100 Ah LiFePO4 battery	Provide electrical backup and night-time supply.
Manage PV energy	MPPT	Charge the battery from the PV array efficiently.
Create cold	R290 vapour-compression refrigeration	Remove heat from the cold chamber and charge the PCM.
Reject heat	Air-cooled condenser + fan	Reject heat to ambient air.
Expand refrigerant	Matched capillary tube	Provide the pressure drop before the evaporator.
Absorb chamber heat	Finned evaporator + circulation fan	Cool and circulate air.
Store thermal capacity	PCM cassette/modules	Provide thermal buffering during interruptions.
Maintain environment	PUF insulated chamber	Reduce heat gain and stabilize temperature.
Sense and control	ESP32 + sensors	Monitor temperature/RH/power/door/compressor and drive control/alerts.

Monitor storage	ESP32-CAM	Estimate crate occupancy / visual condition as a future smart feature.
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STAGE 1 — Problem Framing & Smart-System Requirements

STATUS — Completed as a concept-definition stage; implementation is pending.

Source in handwritten record: pp. 1–3.

1.1 Original design intent

- Provide temporary temperature-controlled storage between harvest and market rather than a permanent multi-week warehouse.
- Focus on the harvest → temporary storage → market link.
- Design around decentralized use at village collection centres, FPOs, cooperatives or farm-gate aggregation points.
- Support vegetables/horticultural produce that can share a moderate-temperature operating regime.
- Provide monitoring and operational information rather than only a refrigeration box.

1.2 Smart features identified

- Grouping vegetables with similar required temperature.
- Identifying the vegetable / produce category and its ripening/storage stage.
- Recording or estimating how long produce has been kept in the chamber.
- Temperature monitoring.
- Humidity monitoring.
- Storage monitoring / approximate occupancy.
- Power-consumption monitoring.
- Backup-duration awareness.
- Alerts for abnormal temperature, humidity, power or door conditions.

1.3 Smart-system philosophy

- The system is not 'smart' merely because it uses an ESP32.
- Closed-loop temperature control: sensor → controller → refrigeration → feedback.
- Energy-aware operation: solar availability and battery state influence operation.
- Thermal-backup management: PCM provides thermal buffering during refrigeration interruptions.
- Predictive-fault indication: temperature trend and compressor status can indicate cooling problems.
- Local/remote monitoring: ESP32 dashboard provides operational information.
- Storage monitoring: camera and door sensor can provide occupancy/usage information.

Decision: The smart layer is treated as a supervisory/control system around the physical cold-storage system, not as a substitute for correct refrigeration and thermal design.

STAGE 2 — Crop Selection & Chamber Capacity

STATUS — Design baseline selected: moderate-temperature crop group, ≈200 kg module.

Source in handwritten record: pp. 2–5.

2.1 Crop strategy

The notes move away from trying to make one fixed-temperature chamber suitable for every vegetable. The prototype is optimized around a moderate-temperature group, initially using tomato, with compatible crops such as cucumber, French beans and chilli.

- Initial moderate-temperature group: tomato, cucumber, French beans, chilli.
- Cabbage/cauliflower and other lower-temperature vegetables were separated conceptually because forcing one temperature regime creates unnecessary compromises.
- Future architecture can be modular: a moderate-temperature module and, potentially, a lower-temperature module.

2.2 Storage-use case

- Short-term aggregation and buffering, not long-duration warehouse storage.
- Normal operation target discussed in the notes: about 24 h of controlled operation.
- Emergency thermal-backup target discussed: initially about 6–12 h, with a longer-term stretch goal of 12–24 h depending on validated PCM quantity and actual heat load.

2.3 Capacity selection

Nominal capacity	Use case in notes
50 kg	Small batch / individual farmer
100 kg	Small collection
200 kg	Medium collection / village collection point
300 kg	Larger village collection
500 kg+	Large commercial collection

Decision: The prototype baseline is ≈200 kg because it is large enough to represent a collection-centre use case while remaining more practical for a first physical prototype. The architecture is intended to scale by adding modules.

2.4 Chamber volume

- Initial internal chamber concept: $1.0\text{ m} \times 1.2\text{ m} \times 1.2\text{ m} = 1.44\text{ m}^3$.
- The notes also identify a practical internal-volume range of roughly $1.2\text{--}1.5\text{ m}^3$ for the 200 kg design.
- The actual required volume must account for crates, airflow, evaporator/PCM space, loading/unloading clearance and insulation.

The handwritten record cites an approximately 20 kg crate assumption and uses $200\text{ kg} \approx 10$ crate-equivalents. The exact crate dimensions and packing density remain to be physically validated.

STAGE 3 — Thermal Load & Cooling Capacity

STATUS — First-order thermal-load model completed; detailed heat-load validation remains.

Source in handwritten record: pp. 4–7.

3.1 Initial pull-down calculation

Baseline assumption recorded in the notebook:

- Produce mass: 200 kg.
- Incoming produce temperature: $\approx 30^{\circ}\text{C}$ — Engineering Assumption.
- Target chamber/produce temperature: $\approx 10^{\circ}\text{C}$ — Prototype Target.
- Temperature difference: $\Delta T \approx 20\text{ K}$.
- Tomato specific heat used: $C_p \approx 3.99\text{ kJ/kg}\cdot\text{K}$.

First-order sensible cooling energy:

$$Q = m \times C_p \times \Delta T = 200 \times 3.99 \times 20 \approx 15,960\text{ kJ} \approx 4.43\text{ kWh thermal}$$

The handwritten page writes 15,960 kJ and also shows a '443 kW' conversion. The later pages consistently treat the value as about 4.43 kWh of thermal energy. This document therefore records the 15,960 kJ result and flags the unit conversion as needing confirmation rather than silently treating '443 kW' as correct.

3.2 Other heat-load terms used in the design

Load term	Working value in notes	Classification
Initial product pull-down	$\approx 4.43\text{ kWh thermal}$	Calculated
Wall / envelope heat gain	$\approx 1.0\text{--}1.9\text{ kWh/day}$ in working estimate	Engineering estimate / requires detailed $U\cdot A\cdot\Delta T$ calculation
Respiration	$\approx 0.07\text{ kWh/day}$ in working estimate	Engineering estimate
Infiltration / door allowance	$\approx 0.42\text{ kWh/day}$ in working estimate	Engineering estimate
Electronics / auxiliary losses	$\approx 0.15\text{ kWh/day}$ in working estimate	Engineering estimate
Total working daily thermal requirement	$\approx 6.3\text{ kWh thermal/day}$	Design estimate

3.3 Refrigeration architecture derived from the load

- Solar PV $\rightarrow$ electrical storage / power management $\rightarrow$ compressor $\rightarrow$ condenser $\rightarrow$ expansion device $\rightarrow$ evaporator $\rightarrow$ cold chamber.
- Target design cooling capacity recorded in the notes: approximately 600 W thermal.
- Preliminary evaporator target: approximately $2\text{--}5^{\circ}\text{C}$ evaporating condition for a $\approx 10^{\circ}\text{C}$ chamber.
- The refrigeration system is sized to pull down the chamber and periodically recharge the PCM rather than continuously running at full load.

Decision: The system should be treated as a hybrid electrical + thermal energy-storage system: the battery stores electricity, while the PCM stores cooling/thermal capacity.

STAGE 4 — PCM & Refrigeration System Design

STATUS — Architecture selected; component sizing and final matching still require datasheet/technician validation.

Source in handwritten record: pp. 8–14.

4.1 PCM design questions

1. What PCM type and commercial material should be used?
2. What phase-change temperature is appropriate?
3. How much PCM is required?

4. Where should PCM be installed?
5. How many hours of realistic backup can it provide?

4.2 PCM baseline

Parameter	Current project value	Status
PCM quantity	≈25 kg	Prototype Target / Engineering Assumption
PCM product notation in notebook	IP08 / written as ZP08 in some pages	Needs final commercial identification
Target phase-change region	≈8–12°C system context	Design requirement; exact PCM must be verified
Packaging	Multiple leak-proof H/OPE/LOPE-type containers / cassette concept	Prototype concept
Placement	Along airflow path / around upper-side or ceiling region in the chamber	Prototype concept
Primary role	Thermal buffer / temperature stabilization	Design Decision

A working energy estimate in the notes uses latent heat $L \approx 180 \text{ kJ/kg}$. At that value, 25 kg corresponds to $\approx 4,500 \text{ kJ} \approx 1.25 \text{ kWh}$ of latent thermal storage. The earlier note also uses 20 kg for $\approx 1 \text{ kWh}$. Actual usable backup will be lower/different depending on the selected PCM's latent heat, phase-change range, sensible heat, heat-transfer rate, state of charge and chamber heat load.

4.3 PCM operating modes

- Daytime / charging: solar-powered refrigeration cools the produce/chamber and charges the PCM.
- Normal target: compressor cycles as needed while the chamber remains within its control band.
- Night: solar is unavailable; battery supplies the refrigeration/control load while PCM remains charged.
- Power failure: compressor stops, ESP32 detects the failure and alerts the operator; PCM absorbs incoming heat and slows temperature rise.
- The design intentionally uses PCM as a thermal battery, not as the primary source of cooling for initial product pull-down.

4.4 Refrigeration architecture

Subsystem	Current selection / concept	Status
Refrigeration cycle	Vapour-compression	Selected
Refrigerant	R290 (propane)	Selected in concept; safety/compatibility must be verified
Condensing unit	Air-cooled R290 condensing unit; notebook references MTH UCSHU 28A SA and Embraco EMT6165U	Provisional / verify exact model
Compressor	Notebook contains conflicting preliminary ratings ($\approx 1/4 \text{ HP}$ and $\approx 0.5\text{--}0.75 \text{ HP}$ class)	Needs final sizing
Condenser	Air-cooled, external, fan-cooled	Selected concept
Expansion device	Matched capillary tube; TXV considered as an alternative	Capillary selected for prototype simplicity
Evaporator	Finned evaporator coil + circulation fan	Selected concept; final capacity/dimensions pending
Target cooling capacity	≈600 W thermal working design value	Prototype target

The project deliberately avoids DIY refrigerant charging. R290 is flammable. The sealed condensing unit, tubing, capillary sizing, leak testing, evacuation and refrigerant charging should be handled by a qualified refrigeration technician using appropriate procedures.

Decision: The current baseline is a conventional sealed R290 vapour-compression refrigeration loop combined with a PCM thermal battery. Peltier cooling is no longer the primary cooling architecture.

STAGE 5 — Solar PV + Battery System

STATUS — Baseline energy architecture selected: $\approx 1.5\text{--}1.65\text{ kWp PV} + 48\text{ V } 100\text{ Ah LiFePO}_4$.

Source in handwritten record: pp. 15–19.

5.1 Electrical energy estimate

- Working thermal requirement: $\approx 6.3\text{ kWh thermal/day}$.
- Working electrical design allowance: $\approx 4\text{ kWh/day}$ for refrigeration + electronics after the project's evolving assumptions.
- Using a 600 W effective refrigeration design value, the notes estimate roughly $10\text{--}11\text{ h}$ equivalent compressor operation on the first pull-down/day.
- At $\approx 300\text{ W}$ electrical compressor input, 10.5 h would be $\approx 3.15\text{ kWh electrical}$; the notes then allow approximately $0.5\text{--}0.8\text{ kWh/day}$ for other loads.
- Working design electrical load adopted in the notebook: $\approx 4\text{ kWh/day}$.

5.2 Solar sizing

Working equation: $\text{PV size} \approx \text{Daily energy} / (\text{peak-sun-hours} \times \text{system efficiency})$

- Daily design energy: $\approx 4\text{ kWh}$.
- Effective solar hours used: $\approx 4.5\text{ h/day}$.
- System efficiency assumption: $\approx 75\%$.
- Calculated minimum: $\approx 1.19\text{ kWp}$.
- Practical selected target: $\approx 1.5\text{ kWp}$.
- Notebook option: $3 \times 550\text{ W panels} = 1.65\text{ kWp}$.

5.3 Battery sizing

- Battery: $48\text{ V}, 100\text{ Ah LiFePO}_4$.
- Nominal stored energy: $48 \times 100 = 4,800\text{ Wh} = 4.8\text{ kWh}$.
- At 80% usable capacity: $\approx 3.84\text{ kWh}$.
- At an assumed 90% conversion efficiency: $\approx 3.46\text{ kWh}$ usable AC-side energy.
- At a 300 W average load, the simple calculation gives $\approx 11.5\text{ h}$, but actual runtime will depend on compressor cycling, startup, auxiliary loads, inverter efficiency and battery limits.

5.4 Power architecture

- Solar PV $\rightarrow$ MPPT $\rightarrow 48\text{ V } 100\text{ Ah LiFePO}_4 \rightarrow$ pure-sine inverter $\rightarrow 230\text{ V AC bus}$.
- The AC bus supplies the refrigeration condensing unit and relevant AC loads.
- A separate DC supply/regulation stage is required for ESP32/sensors and low-voltage electronics.
- Battery and PV protection are required; exact breaker/fuse/isolator ratings must be finalized from real equipment data.

Decision: Current electrical baseline: $3 \times 550\text{ W PV panels}$ ($\approx 1.65\text{ kWp}$), $48\text{ V } 100\text{ Ah LiFePO}_4$ battery, $\approx 48\text{ V}/50\text{ A MPPT}$, $\approx 2\text{ kVA}$ pure-sine inverter with adequate surge capability.

STAGE 6 — Mechanical Chamber Design

STATUS — Mechanical concept defined; fabrication drawings and material selection remain to be finalized.

Source in handwritten record: pp. 20–25.

6.1 Chamber geometry

Parameter	Current design baseline
Internal chamber	$\approx 1.2 \text{ m} \times 1.2 \text{ m} \times 1.0 \text{ m}$ ($\approx 1.44 \text{ m}^3$) in final notebook table; earlier notes also use $1.0 \times 1.2 \times 1.2 \text{ m}$
Insulation	$\approx 100 \text{ mm}$ PUF
Approximate U-value	$\approx 0.22\text{--}0.26 \text{ W/m}^2\cdot\text{K}$ in the notebook estimate
Outer skin	GI / PPGL / coated sheet concept
Inner lining	Food-safe coated surface
Joints	Cam-lock / tongue-and-groove concept
Sealing	Food-safe silicone / gasket
Floor	Insulated and reinforced, water-resistant
Door	Single insulated hinged door; $\approx 700 \text{ mm W} \times 900 \text{ mm H}$ noted
Storage	Ventilated food-grade crates

The notebook contains two closely related chamber dimension conventions: $1.0 \times 1.2 \times 1.2 \text{ m}$ and $1.2 \times 1.2 \times 1.0 \text{ m}$. They describe the same 1.44 m^3 volume; the final fabrication orientation can be chosen based on the actual crate and evaporator layout.

6.2 Internal arrangement

- Approx. 10–12 crates are envisioned for the 200 kg layout, depending on actual crate size and loading.
- Evaporator is placed high on the rear or upper side wall.
- Forced circulation moves cold air from the evaporator through the crate region and back to the evaporator.
- PCM cassette/modules are placed where airflow can interact with a large surface area while avoiding obstruction of produce loading.
- Multiple thinner PCM containers are preferred over one thick block to improve heat-transfer area.
- Drain pan and drain pipe are located below the evaporator for condensate management.
- Electronics compartment is outside the cold chamber and protected from moisture.

6.3 Temperature sensing arrangement

- T1: upper chamber.
- T2: middle/produce zone.
- T3: lower chamber.
- Optional T4: PCM temperature/status.
- Temperature spread = $T_{\text{max}} - T_{\text{min}}$.
- Average chamber temperature = $(T1 + T2 + T3) / 3$.

Decision: The physical design prioritizes air circulation, food-safe surfaces, insulation continuity, condensate management, serviceability and separation of wet/cold space from electronics.

STAGE 7 — Smart Electronics & Control

STATUS — Control concept defined; firmware and electrical implementation are next engineering tasks.

Source in handwritten record: pp. 26–29.

7.1 ESP32 responsibilities

- Temperature monitoring.
- Humidity monitoring.
- Compressor control signal.
- Evaporator/circulation fan control.
- Condenser-fan status monitoring.
- Battery/power monitoring.
- Door monitoring.
- PCM state estimation.
- OLED/local display.
- Wi-Fi dashboard.
- Alerts.
- Data logging.
- Storage/camera interface.

7.2 Sensors

Device	Purpose	Quantity in notes
DS18B20	Primary chamber temperature sensing	3–4
DHT22	Temperature + relative humidity	1
INA219	Voltage/current/power monitoring for a DC electrical path	1
Magnetic reed switch	Door open/closed detection	1
ESP32-CAM	Storage/occupancy/visual monitoring	1
Battery/BMS data	Battery state information	As available / interface-dependent
Compressor current sensor	Compressor status / abnormal operation indication	1

7.3 Control thresholds in notebook

Condition	Working value	Purpose
Default chamber target	$\approx 10^{\circ}\text{C}$	Nominal operating point
Compressor ON example	$T_2 \geq 11^{\circ}\text{C}$	Start cooling when temperature rises
Compressor OFF example	$T_2 \leq 9^{\circ}\text{C}$	Stop cooling after recovery
High-temperature alert	$\approx 13^{\circ}\text{C}$	Operator warning
Adjustable operating range	$\approx 8\text{--}13^{\circ}\text{C}$	Prototype flexibility
RH target	$\approx 90\text{--}95\% \text{ RH}$	Working target in final notebook; earlier notes use 85–95%
Door warning	$\approx 30 \text{ s open}$	Reduce unnecessary heat/moisture ingress
Compressor anti-short-cycle	$\approx 3 \text{ min minimum OFF/ON intervals}$	Protect compressor from rapid cycling

The humidity target appears as both 85–95% and $\approx 90\text{--}95\%$ in different handwritten pages. Treat $\approx 90\text{--}95\%$ as the current notebook baseline, with final crop-specific RH limits to be set after selecting the exact produce mix.

7.4 Energy-saving mode

- When battery SOC becomes low, prioritize temperature preservation and critical sensors/communication.
- Reduce camera frequency, dashboard update frequency, unnecessary fans and other non-critical loads.

- Use PCM as a thermal buffer so the system can avoid unnecessary compressor operation when the chamber is already within its target range.

7.5 Door monitoring

- Closed → normal operation.
- Open → display warning.
- If open for ≈ 30 s → warning/alert.
- Prolonged open events should be logged because they contribute to infiltration heat load.

STAGE 8 — System Integration

STATUS — System-level architecture defined.

Source in handwritten record: pp. 30–37.

8.1 Four interconnected subsystems

6. Solar power generation.
7. Battery + power management.
8. Refrigeration + PCM.
9. ESP32 control + monitoring.

8.2 Physical zones

Zone	Contents
Zone A — Cold chamber	Produce crates, evaporator/fan, PCM modules, temperature/RH sensors, drain system.
Zone B — Refrigeration	R290 condensing unit, compressor, condenser, condenser fan, filter/drier, refrigerant lines, refrigeration controls.
Zone C — Power	PV input, MPPT, LiFePO4 battery, DC protection, inverter, AC protection.
Zone D — Electronics	ESP32, sensor interfaces, display, relay/contactors control, monitoring electronics, protected enclosure.

8.3 Power path

PV → MPPT → 48 V battery → inverter / DC regulation → loads.

The ESP32 should issue control signals through an appropriate isolated driver/relay/contactors arrangement rather than switching a refrigeration compressor directly from a microcontroller GPIO.

8.4 Electrical protection concept

- PV side: DC isolator and overcurrent/surge protection as required by final PV configuration.
- Battery side: BMS, DC fuse/breaker, isolation and appropriate cabling.
- AC side: inverter output protection, MCB and RCD/RCBO as appropriate.
- Electronics: regulated low-voltage supply, fuse/protection and moisture-protected enclosure.
- Compressor control: ESP32 → isolated driver/MOSFET/relay/contactors → compressor control path, with anti-short-cycle logic.

Exact protection ratings must be selected from the final panel string voltage/current, battery manufacturer specifications, inverter manual, compressor nameplate current, cable lengths and applicable electrical standards. The notebook intentionally records protection as 'as required' rather than locking in unverified ratings.

STAGE 9 — Testing & Validation

STATUS — Test plan completed; tests are not claimed as completed.

Source in handwritten record: p. 38.

9.1 Validation objectives

- Cooling performance.
- Temperature uniformity.
- PCM performance.
- Energy performance.
- Solar/battery performance.
- Smart-system performance.

9.2 Planned tests

Test	Procedure / question	Primary measurements
1. Empty-chamber pull-down	Run refrigeration before loading produce.	Time to target; compressor duty cycle; power.
2. Temperature uniformity	Measure upper/middle/lower chamber temperatures.	T1, T2, T3, average and temperature spread.
3. Crate loading	Test 50 kg, 100 kg and 200 kg loading conditions.	Pull-down time, temperature recovery, airflow effects.
4. 200 kg product pull-down	Load the full design mass and measure the full transient.	Product/chamber temperatures, energy used, time.
5. Door-opening test	Repeat controlled door opening/closing events.	Temperature rise and recovery time.
6. PCM charging	Charge the ≈25 kg PCM system.	PCM temperature, charging time, compressor energy.
7. PCM power-failure test	Switch refrigeration OFF and observe thermal backup.	Temperature rise vs time; backup duration.
8. Battery backup	Run representative night-time / low-solar load.	Battery SOC, voltage, current, runtime.

9.3 Acceptance data to record

- Ambient temperature and humidity.
- Initial produce temperature and mass.
- Chamber temperature at multiple positions.
- RH.
- Compressor current/power and runtime.
- PV voltage/current/power.
- Battery voltage/current/SOC.
- PCM temperature and estimated state.
- Door events.
- Time to reach target and time to recover after disturbances.

Decision: No final claim such as 'X hours of backup' should be made until Test 7 is physically performed under a defined heat-load condition.

STAGE 10 — Final Integrated Plan

STATUS — Current project architecture / design baseline.

Source in handwritten record: pp. 39–40.

10.1 Final architecture

Layer	Final baseline
Energy generation	≈1.65 kWp solar PV (3 × 550 W notebook option)
Energy management	≈48 V / 50 A MPPT concept
Electrical storage	48 V, 100 Ah LiFePO ₄
AC conversion	≈2 kVA pure-sine inverter; surge capability to be verified
Refrigeration	R290 vapour-compression system
Compressor/condensing unit	Embraco / MTH components recorded in notebook; final model matching to be verified
Expansion	Matched capillary system
Evaporator	Finned evaporator + forced circulation fan
Thermal storage	≈25 kg PCM, modular cassette/container arrangement
Cold chamber	≈1.44 m ³ internal volume, ≈100 mm PUF insulation
Produce capacity	≈200 kg
Nominal chamber target	≈10°C, adjustable
Humidity	≈90–95% RH working target
Control	ESP32 closed-loop monitoring/control
Monitoring	OLED + Wi-Fi dashboard + alerts + logging
Storage monitoring	ESP32-CAM concept for occupancy/visual monitoring

10.2 Core explanation / pitch logic

Subsystem	Simple explanation
Refrigeration	Creates the cold.
PCM	Stores cold / provides thermal buffering.
Insulation	Prevents cold loss.
Battery	Stores electrical energy.
Solar	Supplies renewable electrical energy.
ESP32	Coordinates monitoring, control and alerts.

The project evolved from Peltier → battery → short-duration cooling into Solar → LiFePO₄ → vapour-compression refrigeration + PCM → controlled cold storage. This evolution is central to the project's engineering story: refrigeration provides active cooling, PCM provides thermal storage, insulation reduces losses, the battery stores electrical energy, solar supplies energy, and the ESP32 coordinates the system.

11. Research & Engineering Findings Consolidated

This section consolidates the research direction developed during the project rather than presenting a new literature review. Exact source citations should be attached to the team's formal bibliography when the SIH submission requires them.

11.1 Problem / field context

- The design is aimed at short-term harvest-to-market buffering in the NER context, where decentralized collection, power reliability and transport constraints make a local cold-storage buffer useful.
- The system is intentionally modular and decentralized rather than a large centralized cold store.
- The target use case is a village collection point / FPO / cooperative / farm-gate aggregation location.
- The chamber is intended to reduce the need to immediately move all harvested produce to market when transport or market timing is inconvenient.

11.2 Crop/storage research direction

- The project moved away from a one-temperature-for-all approach.
- The prototype is centered on moderate-temperature vegetables, initially tomato.
- Target chamber operation is adjustable, with $\approx 10^{\circ}\text{C}$ as the nominal working point in the current design.
- Humidity is treated as an important control variable rather than an afterthought.
- Future modularity can separate moderate-temperature and lower-temperature crop groups.

11.3 Cooling technology comparison that led to the current architecture

Approach	Project-level observation	Current role
Peltier / TEC	Simple for small prototypes but poor fit for a 200 kg, long-duration agricultural chamber.	Rejected as primary cooling.
Conventional vapour compression	Scalable, established, appropriate for higher cooling loads.	Primary active cooling.
PCM-only cooling	Useful as thermal storage but requires a way to recharge the PCM and does not replace initial active cooling.	Thermal backup/storage.
Hybrid refrigeration + PCM	Combines active cooling with thermal buffering and can reduce dependence on continuous compressor operation.	Selected architecture.

11.4 PCM research direction

- The project investigated commercial PCM solutions suitable for the vegetable cold-chain temperature range.
- Commercial references considered included Pluss savE®-type products and other PCM suppliers such as Tessol and PC Mwala.
- The notebook's current PCM entry is approximately 25 kg of an IP08/ZP08-labelled material; the exact product name and datasheet must be frozen before procurement.
- The PCM should be selected by actual melting range, latent heat, density, packaging compatibility, heat-transfer rate and safety — not simply by the largest kJ/kg number.

11.5 Important research discipline

- Do not quote a generic vegetable storage temperature as if it applies to every crop.
- Do not claim a specific PCM backup duration without defining the heat load and measuring it.

- Do not claim the refrigeration system is '600 W' merely because the first-order design target is 600 W; the actual delivered capacity depends on evaporating/condensing conditions and equipment selection.
- Do not treat the 200 kg capacity as a proven commercial throughput; it is the prototype design capacity.

12. Master Preliminary Bill of Materials

Compiled from the handwritten Stage 7–10 lists. Quantities are preliminary.

Subsystem	Component	Qty / baseline	Status / note
Mechanical	100 mm PUF insulated panels	As required	Final dimensions/fabrication drawing required
Mechanical	Insulated door + gasket	1	Approx. 700 × 900 mm noted
Mechanical	Food-grade crates	≈10–12	Actual crate dimensions to be frozen
PCM	PCM cassette/module	≈10 × 2.5 kg concept	Exact commercial PCM to be verified
PCM	H/OPE/LOPE leak-proof containers	10 concept	Packaging compatibility required
Refrigeration	R290 condensing unit	1	Exact model/rating to be verified
Refrigeration	Finned evaporator	1	≈700–800 W concept
Refrigeration	Evaporator circulation fan	1+	Airflow sizing required
Refrigeration	Refrigerant tubing	As required	Technician sizing
Refrigeration	Matched capillary	1	Must be matched to final system
Refrigeration	Filter drier	1	As required
Solar	550 W PV panel	3	≈1.65 kWp total
Solar	48 V / 50 A MPPT	1	Final PV/battery compatibility check
Solar	48 V 100 Ah LiFePO4 battery	1	4.8 kWh nominal
Solar	≈2 kVA pure-sine inverter	1	Surge capability must match compressor
Protection	DC isolator/fuses/breakers	As required	Final electrical design
Protection	AC MCB + RCD/RCBO	As required	Final electrical design
Electronics	ESP32 development board	1	Main controller
Electronics	DS18B20	4 concept	3 chamber + optional PCM
Electronics	DHT22	1	RH + temperature
Electronics	INA219	1	DC voltage/current/power path
Electronics	Magnetic reed switch	1	Door
Electronics	ESP32-CAM	1	Occupancy/visual monitoring concept
Electronics	OLED I ² C display	1	Local dashboard
Electronics	Relay/contacter/isolated driver	1+	Compressor control path
Electronics	Emergency stop	1	Safety
Electronics	Electrical enclosure	1	Moisture-protected external enclosure

13. Major Decision Log

Decision	Current choice	Reason / project logic
Prototype capacity	≈200 kg	Represents a medium village collection-point scale and remains practical for prototyping.

Crop strategy	Moderate-temperature group	Avoids trying to optimize one chamber for incompatible crop temperature requirements.
Primary crop	Tomato	Used as the current design/reference crop in thermal calculations.
Primary cooling	Vapour compression	Better suited to the required cooling scale than Peltier.
Thermal storage	PCM	Provides thermal buffering and helps during night/power interruptions.
PCM quantity	≈25 kg	Current engineering target; must be validated experimentally.
Refrigerant	R290	Current design choice for the refrigeration loop; safety and equipment compatibility must be verified.
Expansion	Matched capillary	Simple prototype architecture; TXV was considered.
Insulation	≈100 mm PUF	Current thermal-envelope concept.
Energy source	Solar PV	Core problem statement requirement and renewable supply.
Electrical storage	48 V 100 Ah LiFePO4	Large enough working electrical buffer for the current concept.
Controller	ESP32	Low-cost monitoring/control platform with Wi-Fi capability.
Smart monitoring	Temp + RH + power + door + camera	Provides system/environment/storage awareness.

14. What Is Still NOT Proven / Must Be Done Next

These are not failures of the design; they are the normal transition from concept engineering to physical validation.

- Freeze the exact PCM commercial product, datasheet, melting range and latent heat.
- Confirm whether the notebook's 'IP08/ZP08' notation refers to the exact intended commercial PCM product.
- Finalize compressor/condensing-unit model and verify cooling capacity at the intended evaporating and condensing conditions.
- Resolve the notebook's conflicting compressor-size notes (≈1/4 HP vs ≈0.5–0.75 HP class).
- Select/size the evaporator using the actual refrigeration capacity and air-side conditions.
- Match the capillary tube to the final compressor/condenser/evaporator/refrigerant conditions.
- Perform a complete heat-load calculation including conduction, infiltration, respiration, internal loads and safety margin.
- Finalize PV string configuration, MPPT rating and protection based on actual panel electrical characteristics.
- Finalize inverter continuous and surge rating using the actual compressor startup behavior.
- Finalize DC/AC protection and cable sizes from actual currents, lengths and applicable electrical requirements.
- Build the chamber and verify insulation continuity, air leakage and condensate drainage.
- Implement ESP32 firmware and compressor anti-short-cycle protection.
- Calibrate all temperature/RH sensors.
- Run Stage 9 tests and record actual data.

- Only after testing, publish backup duration, cooling time, energy consumption and temperature-uniformity claims.

15. AI HANDOFF — COPY/PASTE CONTEXT

The following block is deliberately written so the team can paste it into another AI tool together with this document.

You are assisting with SIH 2026 Hardware Problem Statement 26005: “Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in the North Eastern Region (NER).”

CURRENT PROJECT BASELINE

- Target use: decentralized short-term harvest-to-market cold-storage buffer at village collection centres / FPOs / cooperatives / farm-gate points.
- Prototype capacity: approximately 200 kg produce.
- Reference crop: tomato; compatible moderate-temperature vegetables include cucumber, French beans and chilli.
- Chamber: approximately 1.44 m³ internal volume, approximately 100 mm PUF insulation.
- Nominal chamber target: approximately 10°C, adjustable.
- Working RH target: approximately 90–95% RH; exact crop-specific limits remain to be finalized.
- Active cooling: R290 vapour-compression refrigeration.
- Condenser: air-cooled external condenser with fan.
- Expansion: matched capillary-tube concept.
- Evaporator: finned evaporator + forced circulation fan.
- Thermal storage: approximately 25 kg PCM in modular/cassette form; exact commercial PCM product must be verified.
- Electrical storage: 48 V, 100 Ah LiFePO₄.
- Solar: approximately 1.65 kWp concept using 3 × 550 W panels.
- MPPT: approximately 48 V / 50 A concept.
- Inverter: approximately 2 kVA pure-sine concept, with compressor surge capability to be verified.
- Controller: ESP32.
- Sensors: 3–4 × DS18B20, DHT22, INA219/power sensing, magnetic door switch, optional compressor-current sensing, ESP32-CAM, OLED.
- Smart functions: closed-loop temperature control, RH monitoring, energy-aware operation, PCM status estimation, door alerts, power/battery monitoring, dashboard, logging, camera/occupancy monitoring.
- Refrigeration control: ESP32 must not directly switch the compressor; use an appropriate isolated driver/relay/contactors/control arrangement and anti-short-cycle logic.
- R290 is flammable. Refrigerant charging and sealed-system work must be performed by a qualified refrigeration technician.

ENGINEERING CLASSIFICATION RULE

Always label important values as one of:

1. Evidence-Based
2. Calculated
3. Engineering Assumption
4. Prototype Target

DO NOT

- Treat an engineering assumption as a measured fact.
- Claim a backup duration without a defined heat load and test.
- Claim 600 W actual refrigeration capacity without equipment/datasheet/test confirmation.
- Treat 25 kg PCM as experimentally proven.
- Restart the project from zero without first using the established design baseline.
- Replace a selected subsystem without explaining what decision it changes and why.

CURRENT NEXT WORK

The project is moving from concept/design into prototype execution and validation. Priority work is:

1. Freeze exact PCM product/specification.

2. Freeze and verify refrigeration equipment sizing.
3. Finalize evaporator/capillary matching.
4. Finalize heat-load and energy model.
5. Finalize electrical protection and wiring.
6. Build/test chamber.
7. Implement ESP32 control and dashboard.
8. Run Stage 9 validation tests.
9. Replace targets with measured results where possible.

When proposing any new design, first state whether it is:

- compatible with the current architecture,
- a modification of the current architecture, or
- a replacement of a current decision.

Use this document as the project's current source of truth unless the team supplies newer validated data.

16. Team Working Checklist

Work area	Owner should produce	Definition of done
Refrigeration	Final compressor/condensing unit + evaporator + capillary sizing	Datasheets, schematic, technician confirmation and safe sealed-system plan
PCM	Exact material, quantity, packaging, placement	Datasheet + calculated capacity + physical charging method
Mechanical	Chamber CAD/drawing + crate layout + insulation details	Manufacturable drawing with service access and drainage
Solar/power	PV/MPPT/battery/inverter single-line diagram	Ratings, protections, cable sizes and load budget
Electronics	ESP32 wiring + sensor map + control logic	Bench-tested firmware and safe output drivers
Software/dashboard	Local/remote monitoring UI	Live values, logging and alerts demonstrated
Testing	Stage 9 test sheets	Raw data recorded with timestamps and conditions
Documentation/PPT	SIH narrative and visuals	Every claim traceable to evidence/calculation/test

17. Handwritten Notebook Source Map

Notebook pages	Content captured in this master document
1–3	Smart features, crop/storage philosophy, short-term storage intent, crop grouping.
4–5	200 kg capacity, crate concept, chamber volume, scalability.
6–7	Thermal load, cooling capacity, PCM energy estimate, refrigeration architecture.
8–14	PCM design, refrigeration loop, R290, compressor/condensing unit, evaporator, capillary, final refrigeration baseline.
15–19	Solar sizing, daily electrical energy, 48 V 100 Ah LiFePO4, MPPT, inverter, power architecture.
20–25	Mechanical chamber, insulation, door, crate arrangement, airflow, PCM placement, sensors and final dimensions.
26–29	ESP32 control, sensors, thresholds, energy-saving mode, dashboard, alerts, camera monitoring.
30–37	Final electronics architecture, electronics BOM, system integration, protection and control path.
38	Stage 9 testing and validation plan.
39–40	Stage 10 final integrated architecture and project evolution/pitch logic.

18. Final Project Snapshot

The project is now best understood as a modular, solar-powered, smart, short-term agricultural cold-storage unit. It is not simply a solar refrigerator and not simply a PCM box. Its architecture combines four forms of buffering/control: refrigeration removes heat, PCM stores thermal capacity, the battery stores electrical energy, and insulation slows heat ingress. The ESP32 supervises the system and makes the physical hardware observable and controllable.

The major engineering transition now is from 'designing the concept' to 'proving the concept'. The next strongest evidence for the project will therefore come from a working prototype, measured cooling/energy/PCM data, and a clean trace from each SIH claim to a calculation, datasheet, research source or test result.

Compiled from the team's supplied 40-page handwritten SIH 26005 record and the established project work in this conversation. This document intentionally does not claim that the Stage 9 tests have already been performed.